MICHAL HORNICKY

SOFTWARE ENGINEER

CONTACT

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DATE OF BIRTH

24 January 1995

ABOUT ME

Personal computing, Internet and Smartphones were the last global paradigm shifts. I want to work on the next one

EXPERIENCE

LLMs

Machine Learning Infrastructure

Distributed systems

Cloud

Web

Android

GPGPU programming

WORK EXPERIENCE

SOFTWARE ENGINEER, GOOGLE

Zurich | November 2021 - Current

- Developed tooling, testing processes and automated testing architecture for Google Assistant, Google Search, and Gemini

SENIOR SOFTWARE ENGINEER, REDHAT

Brno | July 2020 - September 2021

Developed on the Patch application on cloud.redhat.com

- Led the backend development for the Patch application
- Scaled the patch application to serve thousands of large scale deployments per day.
- Helped other teams with developing and deploying their applications on the same platform.

SOFTWARE ENGINEER, REDHAT

Brno | September 2019 - July 2020

Primarily worked on distributed web application back-ends deployed on a streaming architecture using OpenShift Kubernetes.

- Continually optimized critical part of an application performing continuous vulnerability detection on cloud.redhat.com
- Initiated the development of the Patch application in Go, which was performing continuous update detection on customers systems.
- Developed libraries and tools for integrating the Go language into pre-existing python based ecosystem.

ANDROID DEVELOPER, TOUCHART S.R.O

Brno | March 2016 - September 2019

I worked in a small team developing Android applications for business customers.

- Managed all aspects of several Android applications, from initial research, customer communication, development, testing, and deployment.
- Spearheaded architectural overhauls, migration to Kotlin, usage of MVVM and Reactive dataflow paradigms.
- Developed large amount internal libraries

LANGUAGES

Rust

Go

Python

Kotlin - Java

C++, C

Javascript - React

EDUCATION

MASTERS DEGREE - INFORMATION TECHNOLOGY,
Brno university of technology | 2017-2019

BACHELORS DEGREE - INFORMATION TECHNOLOGY,
Brno university of technology | 2014-2017

NOTABLE PROJECTS

MML | GITHUB

Minimalist machine learning library in rust.

- Small library for ML training & inference on Apple Mx devices using both CPU and GPU.
- Written in rust, intended for performing vision related tasks for video editing.

BLOK3 | WEB

Low-latency cryptorcurrency portfolio provider.

- Built using self-developed Stor & Protokit libraries.
- Provides extremely detailed cryptocurrency portfolio information
- Built for low-latency operation, even without any form of caching.

PROTOKIT | GITHUB

High-performance implementation of protocol buffers for Rust.

- Only Rust implementation that supports full specification of the TextFormat encoding of protocol buffers.
- Competitive performance with the widely used ‘prost‘ library, with larger feature set

STOR | GITHUB

High-performance, low-latency embedded key-value database.

- Built on RocksDB
- provides a typed key-value API.
- Critical piece of infrastructure, required to build Blok3.

LNPAY | GITHUB

Payment platform, allowing user to buy Bitcoin on the lightning network using credit cards.

- Connected to credit card network using the Stripe platform
- Directly communicated with the lightning network
- Used an on-exchange hedging strategy to reduce risk of unfinished transactions

- Written in Go and deployed on Azure cloud using Kubernetes

PROSPECTOR | BITCOINTALK

A green-field implementation of a cryptocurrency miner.

- Written in Rust, used OpenCL and Cuda for GPU computing
- Achieved worlds best performance on Signatum cryptocurrency
- Highly extensible, supported both CPU and GPU mining, 6 different cryptocurrency mining algorithms

CRYPTRADER | GITHUB

Distributed application for automated day-trading of cryptocurrencies. Developed as part of master's thesis.

- Written in Rust programming language, using Kubernetes as a deployment platform
- Event-based architecture built on ZeroMQ and NATS
- Was never deployed commercially due to issues with trading approach (90% of all day-traders lose money)